

Exercise Set 1

Quantitative Trading and Price Impact
MSc in Mathematics and Finance

2025-2026

Download, unzip, and load the binned sample data. You may find the Python libraries `zipfile` and `pandas` useful.

- (a) **Computing volume curves.** A key *non-directional* intraday feature is the volume curve

$$V_t = \mathbb{E} \left[\sum_{s>t} |v_s| \middle| \mathcal{F}_t \right],$$

which measures the expected *remaining* traded volume from time t until the market close.

In this exercise, estimate V_t empirically as follows: for each symbol and each trading day, compute the cumulative volume traded *after* each intraday time bin t . Then, for each t , average this remaining volume across the previous month of trading days. Concretely, the value at 12:00 should equal the average (over the last month) of the total volume traded after 12:00 on each day. Compute this volume curve on the full binned dataset (all symbols, all dates).

Hint: Your result should be one volume curve per (date, stock) pair, e.g. stored as a DataFrame with a MultiIndex (date, stock) on the rows and one column per time bin. The `pivot` function from `pandas` can be helpful here.

- (b) **Computing VWAP schedules.** Implement each of the following three VWAP-style execution schedules. The inputs are the intended total trade size Q , the (signed) traded volume series $(v_t)_t$, and (optionally) the volume curve $(V_t)_t$. The output is the schedule $(q_t)_t$.

Remark: The first two schedules below are *not implementable* in live trading because they rely on full-day information: they assume access to the entire day's realized volume profile $(v_t)_t$ (and/or its sum) at the start of the day. The third schedule replaces this look-ahead with the (pre-estimated) volume curve and can be deployed.

- **Naïve.** Assume no slippage and allow look-ahead by pretending the full intraday volume profile is known at the start of the day:

$$q_t^{\text{naïve}} = \frac{|v_t|}{\sum_s |v_s|} Q.$$

- **Slippage.** Incorporate a latency $\delta t = 10$ s by shifting the volume signal back in time, but still normalize by the full-day total volume:

$$q_t^{\text{slippage}} = \frac{|v_{t-\delta t}|}{\sum_s |v_s|} Q.$$

- **Realistic.** Remove look-ahead by using the volume curve V_t as an estimate of the remaining volume from time t to the close:

$$q_t^{\text{realistic}} = \min \left\{ \frac{|v_t - \delta t|}{V_t}, 1 \right\} \left(Q - \sum_{s < t} q_s^{\text{realistic}} \right).$$

Compute all three schedules for every (date, stock) pair in the dataset, producing a 10-second schedule $(q_t)_t$ for each pair.

- (c) **Creating synthetic alphas.** Define the forward return r^h over horizon $h > 0$ and a synthetic alpha α^h by

$$r_t^h = \frac{P_{t+h} - P_t}{P_t},$$

$$\alpha_t^h = \frac{x(P_{t+h} - P_t) + y(W_{t+h} - W_t)}{P_t} = x r_t^h + y \frac{W_{t+h} - W_t}{P_t},$$

where W is a simulated standard Brownian motion (independent of the price process P) and $x, y \in \mathbb{R}$ are constants.

1. For given values of x and y , derive analytically the variance $\text{Var}(\alpha_t^h)$ and the correlation $\text{Corr}(\alpha_t^h, r_t^h)$.
2. Fix a target correlation ρ . Determine x and y such that

$$\mathbb{E}[r_t^h | \alpha_t^h] = \alpha_t^h \quad \text{and} \quad \text{Corr}(\alpha_t^h, r_t^h) = \rho.$$

(The first condition makes the synthetic alpha α_t^h a conditionally unbiased predictor of the forward return r_t^h , i.e., whenever the signal equals a , the average realized return r_t^h is a , while the second condition separately controls how noisy that prediction is.)

3. Choose a value of ρ and set $h = 10$ s. Using the binned sample data, estimate the quantities needed to instantiate your formulas from part 2, in particular $\text{Var}(r_t^h)$ and $\mathbb{E}[P_t^{-2}]$, as one scalar value per stock by pooling over all dates and intraday time bins. Compute the corresponding coefficients x and y for each stock. Then simulate W , construct α_t^h , and report (i) the empirical variance $\text{Var}(\alpha_t^h)$, and (ii) the empirical correlation $\text{Corr}(\alpha_t^h, r_t^h)$ per stock.